torch.rand#

https://pytorch.org/docs/stable/generated/torch.rand.html#torch.rand

Description:

Returns a tensor filled with random numbers from a uniform distribution on the interval [0,1)[0, 1)[0,1) The shape of the tensor is defined by the variable argument size. size (int...) – a sequence of integers defining the shape of the output tensor. Can be a variable number of arguments or a collection like a list or tuple. generator (torch.Generator, optional) – a pseudorandom number generator for sampling out (Tensor, optional) – the output tensor.

Function Signature

```
torch.rand(*size, *, generator=None, out=None, dtype=None,
layout=torch.strided, device=None, requires_grad=False,
pin_memory=False) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

generator (torch.Generator, optional) – a pseudorandom number generator for sampling out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False. pin_memory (bool, optional) – If set, returne

Code Examples

```
>>> torch.rand(4)
tensor([ 0.5204,  0.2503,  0.3525,  0.5673])
>>> torch.rand(2, 3)
tensor([[ 0.8237,  0.5781,  0.6879],
        [ 0.3816,  0.7249,  0.0998]])
```

Detailed Documentation

Documentation

Returns a tensor filled with random numbers from a uniform distribution on the interval [0,1)[0, 1)[0,1)

The shape of the tensor is defined by the variable argument size.

size (int...) – a sequence of integers defining the shape of the output tensor. Can be a variable number of arguments or a collection like a list or tuple.

generator (torch.Generator, optional) – a pseudorandom number generator for sampling

out (Tensor, optional) – the output tensor.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

pin_memory (bool, optional) – If set, returned tensor would be allocated in the pinned memory. Works only for CPU tensors. Default: False.

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